



Colour in Water

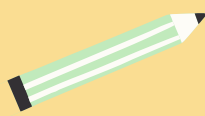
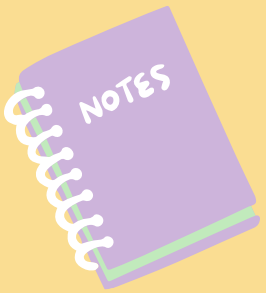
CONVECTION CURRENTS

A Fun Science Adventure!

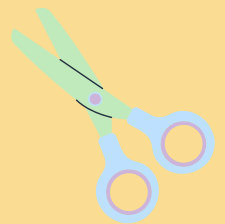




What are Convection Currents?

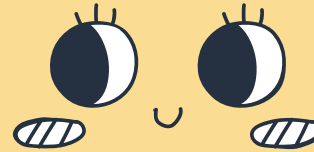


When water gets warm, it moves up. When it cools down, it moves down. This creates a circular movement called convection! It's like a merry-go-round inside your water!





The Colour in Water Experiment



What You Need:

- A clear glass or jar
- Warm water
- Cold water
- Food colouring (any bright colour works!)

Gather all these materials before we start our exciting experiment!



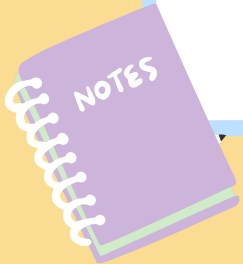
Let's Do the Experiment!



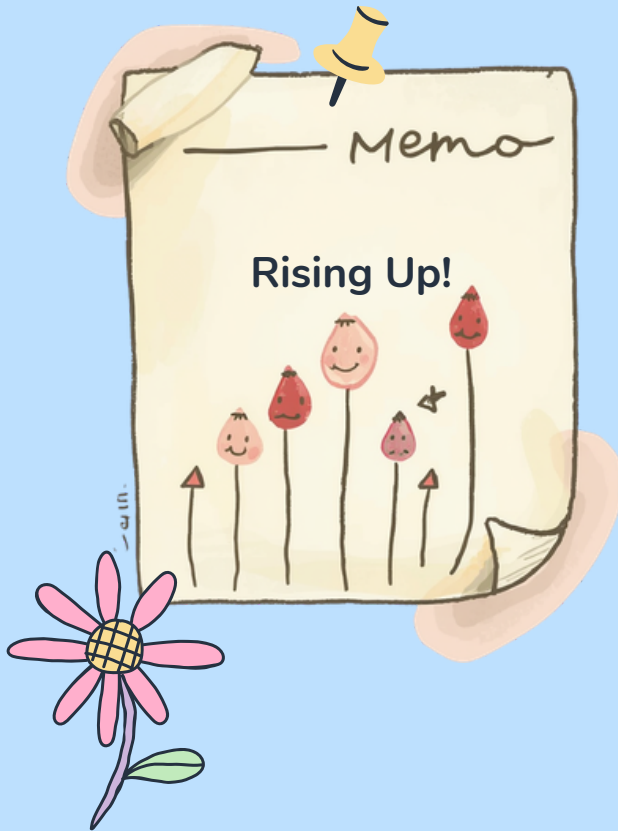
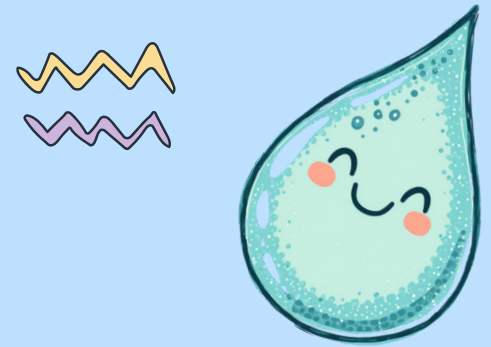
- Step 1: Fill the glass with cold water.
 - Step 2: Add a drop of food colouring to warm water.
 - Step 3: Gently pour the coloured warm water into the glass.
- Watch what happens!



Follow these steps carefully and observe what happens to the coloured water. Remember to watch closely - science is all about observing!



What Do You See?

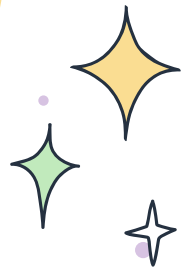


Why Does This Happen?



Warm water is lighter, so it floats up to the top! Cold water is heavier, so it sinks down to the bottom. This movement of water is called a convection current.

The food colouring helps us see this amazing movement happening right before our eyes!





Convection Currents Around Us



Hot air balloons rise up high!
The burner heats the air
inside, making it lighter and
lifting the balloon into the sky.

Watch water boiling in a pot!
Hot water rises from the
bottom while cooler water sinks
down, creating a swirling
dance.

Wind blowing outside is
convection too! The sun
heats the ground, warm
air rises, and cooler air
rushes in.

Ocean currents move around
the world! Warm water near
the equator rises and travels,
while cold water sinks and
flows.

Question 1: What happens to warm water - does it go up or down?



Think and Answer!



Question 2: Why does cold water sink to the bottom?

Question 3: Where else can you see convection currents in real life?



GREAT JOB, YOUNG SCIENTISTS!

You learned how heat makes water move. Keep exploring and asking questions!

